

Manufacturing Test Plan

For all boards (Basic functionality):

Test 1: 5V and 12V output

Connect the input of DPS to a DC power supply. Connect the 5 V output to 250 Ω and the 12 V output to 600 Ω . Set the multimeter to measure the voltage. Use 3.8V as the input voltage. Place the multimeter probes across the 5V load resistor and check whether the output is 5 ± 0.15 V (compare with datasheet). Then repeat the process for the 12V load resistor to see if the output is 12 ± 0.8 V.

Test 2: Short Circuit current <30 mA

Use the power supply to provide an input voltage of 3.8 V to the input pins of the DPS. Connect the digital multimeter in series with the output pins of each 5 V and 12 V output to measure the short-circuit current through each output. Then turn on the power supply to check whether the current reading on the ammeter is below 30mA.

Test 3: The DPS shall maintain voltage output specifications with an input voltage +/-20% from nominal.

1. Use the power supply to provide a nominal input voltage of 3.8 V to the input pins of the DPS.
2. Connect a 250 Ω load to the 5 V output pins and a 600 Ω load to the 12 V output pins.
3. Connect the digital multimeter in parallel with each load to measure the voltage across each load.
4. Connect the digital multimeter in series with each load to measure the current through each load resistor.
5. Vary the 3.8 V input voltage $\pm 20\%$ from nominal in steps (i.e. from 3.04 V to 4.56 V), at each input voltage setting, measure the corresponding 5 V and 12 V output voltages across each load resistor using the oscilloscope.
6. Check whether the 5 V output voltage remains within 4.85 V – 5.15 V and the 12 V output voltage remains within 11.2 V – 12.8 V.

For 1 in every 20 boards

Test 4: Power Efficiency >60%

1. Use the power supply to provide a 3.8 V to the input pins of the DPS.
2. Connect a 250 Ω load to the 5 V output pins and a 600 Ω load to the 12 V output pins.
3. Connect the digital multimeter in parallel with each load to measure the voltage across each load.
4. Connect the digital multimeter in series with each load to measure the current through each load resistor. Turn on the power supply and record the readings.
5. Use the formula below to check whether the efficiency is above 60%:

$$\eta = \frac{(voltage\ across\ 250\Omega \times current\ through\ 250\Omega) + (voltage\ across\ 600\Omega \times current\ through\ 600\Omega)}{input\ voltage \times input\ current}$$

For 1 in every 50 boards

Test 5: The DPS shall not suffer damage at temps down to -40°C.

1. Place the DPS inside a thermal chamber. Gradually decrease the temperature to -40 °C .
2. Use a wide-range scientific thermometer to monitor the temperature inside the chamber. Once -40 °C is reached, carefully take the DPS out using protection gloves or tongs.
3. Visually inspect the DPS for any signs of physical damage
4. Connect the DPS to an input voltage of 3.8 V. Connect a 250 Ω load to the 5 V output pins and a 600 Ω load to the 12 V output pins.
5. Verify basic functionality of the DPS by conducting Test 1, Test 2 and Test 3.
6. Leave the DPS for about an hour or until the DPS returns to room temperature. Then repeat step 5.

Test 6: The DPS shall maintain voltage output specifications over a temperature range of 0-70°C.

1. Place the DPS inside a thermal chamber. Gradually increase the temperature from 0 °C to 70 °C in 10 °C increments.
2. Use a wide-range scientific thermometer to monitor the temperature inside the chamber. At each 10 °C increment, carefully take the DPS out using protection gloves or tongs. Note: Let the DPS inside the thermal chamber to stabilise at each temperature before taking out for measurements.
3. Connect the DPS to an input voltage of 3.8 V. Connect a 250 Ω load to the 5 V output pins and a 600 Ω load to the 12 V output pins.
4. Verify basic functionality of the DPS by conducting Test 1, Test 2 and Test 3. This process must be quick so that the DPS maintains at the desired temperature.
5. Place the DPS back to the thermal chamber and repeat step 3 and 4 at each 10 °C increment.